

24CS31 — Last-Minute Prediction Sheet

Laplace Transforms & Vector Space · SEE 100 marks / 3 hrs · **Answer one full question from each unit** · Q1|Q2 → Unit I Q3|Q4 → II Q5|Q6 → III Q7|Q8 → IV Q9|Q10 → V

THE 60-SECOND RULE. Don't read all ten questions. In each unit, find the **one trigger** below and answer **that** question — the rest of its parts are the ones you studied. **I:** the t^n proof • **II:** the Heaviside piecewise • **III:** the reflection→rotation→contraction question • **IV:** QR + least-squares (that's Q8) • **V:** orthogonally diagonalize.

READ THE TWO RIGHT-HAND COLUMNS SEPARATELY — they mean different things.

“**Topic came up**” = how many of the four papers asked *this kind of question at all*, with any numbers. “**SAME values**” = how many asked it with the *identical matrix, function or constants* printed here. A row reading **4/4 topic / 0/4 same** (e.g. QR factorization) is **certain to appear but the matrix will be new** — you must own the method, not an answer. A row reading **4/4 / 3/4** (orthogonal diagonalization) is a question you can rehearse end-to-end and reproduce. Colour is just the size of the number: **4/4** **3/4** **2/4** **1/4** **0/4**. Proofs count as “same values” when the wording is identical. The **2-mark definition slot only exists in the March 2024 format**, so it can never score above 1/4 — it is still near-certain under the current pattern.

UNIT I — Laplace Transforms

20 marks · a(2) + b(4) + c(7) + d(7)

PICK the question containing the t^n proof. The $\sin \sqrt{t}$ question is **always** in the same one (4/4 papers) — that's 11 marks together. The periodic-function problem is always in the *other* one.

#	Question — write this	Marks	Section	Topic came up	SAME values
1	Prove $L\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} \{F(s)\}$, n a positive integer. ← THE trigger. Do this first.	07	Q1c or Q2b	4/4	4/4
2	Given $L\{\sin \sqrt{t}\} = \frac{\sqrt{\pi}}{2s^{3/2}} e^{-1/4s}$, show $L\{\frac{\cos \sqrt{t}}{\sqrt{t}}\} = \frac{\sqrt{\pi}}{s^{1/2}} e^{-1/4s}$	04	Q1b — with #1	4/4	2/4
3	Evaluate (i) $L\{t e^{-4t} \sin 3t\}$ (ii) $L\{\frac{t - \sinh at}{t}\}$ (multiply by $t \rightarrow -F'(s)$; divide by $t \rightarrow \int_s^\infty F$)	07	Q1d or Q2c	4/4	1/4
4	Evaluate (i) $\int_0^\infty \frac{e^{-t} \sin t}{t} dt$ (ans $\pi/4$) (ii) $L\{t \sin 3t \cos 2t\}$	07	Q1d	3/4	1/4
5	Periodic function. Triangular wave, period $2a$: $f(t) = \begin{cases} t, & 0 \leq t \leq a \\ 2a - t, & a \leq t \leq 2a \end{cases}$ <i>ans</i> $\frac{1}{s^2} \tanh(\frac{as}{2})$ OR half-rectifier $E \sin(\omega t)$ on $(0, \pi/\omega)$, 0 after	07	OTHER question	4/4	2/4
6	Obtain $L\{(\sqrt{t} + 1/\sqrt{t})^3\}$ (expand first, then term-by-term)	04	Q2b	2/4	1/4
7	Definition (2 marks — free). “Write the Laplace transform of a periodic function” = $\frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt$ • or “Define the Laplace transform”	02	Q1a & Q2a	2/4	1/4

UNIT II — Application of Laplace Transforms

20 marks · a(2) + b(4) + c(7) + d(7)

PICK the question containing the Heaviside piecewise function. An ODE problem is **always** sitting next to it (4/4 papers) — that's **14 guaranteed marks** in one question. Convolution is in the other one.

#	Question — write this	Marks	Section	Topic came up	SAME values
1	Express in terms of the Heaviside function and find its LT: $f(t) = \begin{cases} t^2, & 0 < t < 2 \\ 4t, & 2 < t < 4 \\ 8, & t > 4 \end{cases}$ ← THE trigger. Same values twice.	07	Q3c	4/4	2/4
2	Solve by Laplace transforms (one of these, in the <i>same</i> question as #1): • $dx/dt - 2y = \cos 2t$; $dy/dt + 2x = \sin 2t$, $x(0)=1$, $y(0)=0$ • $y'' + 4y' + 3y = e^{-t}$ (set twice, different ICs) • LR circuit: $L di/dt + Ri = Ee^{-at} \rightarrow i = \frac{E}{R-aL} (e^{-at} - e^{-Rt/L})$	07	Q3d / Q4d — with #1	4/4	2/4
3	Convolution theorem. $L^{-1}\{\frac{s^2}{(s^2+a^2)(s^2+b^2)}\}$ OR verify it for $f_1=t$, $f_2=\cos t$	07	OTHER question	4/4	2/4

4	Evaluate $L^{-1}\{\log \frac{s^2+1}{s(s+1)}\}$ differentiate F , then $L^{-1}\{F\} = -t f(t)$	04	Q3b	3/4	2/4
5	Find $L^{-1}\{\frac{3s+2}{s^2-s-2}\}$ — partial fractions. Every paper carries a short inverse LT in <i>both</i> options.	04	Q4b	4/4	1/4
6	Definition (2 marks — free). Define the unit step function + graph • Define the Dirac-delta function + sketch	02	Q3a & Q4a	1/4	1/4

UNIT III — Vector Space and Linear Transformation

20 marks · a(6) + b(7) + c(7)

PICK the question containing the reflection→rotation→contraction problem (usually Q6). In the two newest papers that question also held the transition matrix *and* kernel&range — three 4/4 topics, a clean 20.

#	Question — write this	Marks	Section	Topic came up	SAME values
1	Matrix of a composition + image of a point. Reflection in the x-axis, then rotation $\pi/2$, then contraction of factor 1/3. Find the image of $\begin{bmatrix} 4 \\ 1 \end{bmatrix}$. ← THE trigger. Multiply right to left : $k R_0$ Ref. Reflect in $x = \text{diag}(1, -1)$; in $y = \text{diag}(-1, 1)$.	06	Q6a	4/4	1/4
2	Transition matrix. $B = \{(1, 2), (3, -1)\}$, $B' = \{(3, 1), (5, 2)\}$ of R^2 . Find the transition matrix from B to B' ; if $[u]_B = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, find $[u]_{B'}$. Same bases twice running.	07	Q6b — with #1	4/4	2/4
3	Kernel and range + verify rank-nullity for $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 1 \\ 1 & 1 & 4 \end{bmatrix}$ (rank 2, nullity 1). Be ready for the formula version too, e.g. $T(x, y) = (3x, x-y, y)$.	07	Q6c — with #1, #2	4/4	1/4
4	Prove T is linear + find images. $T: P_2 \rightarrow P_2$, $T(ax^2+bx+c) = (a+b)x^2+c$; image of $5x^2+6x+1$. OR $T(x, y) = (3x+y, 2y, x-y)$; images of $(1, 2)$, $(2, -5)$	07	Q5b — all 4 papers	4/4	2/4
5	Span / linear combination / independence / basis. e.g. is a set of four 2×2 matrices a basis of M_{22} ? (flatten each to a 4-vector, take the 4×4 determinant) • or: does $\{(1, 2, 3), (-1, -1, 0), (2, 5, 4)\}$ span R^3 ?	06	Q5a	4/4	1/4
6	Prove $T(x) = Ax$ rotates x through θ about the origin if $A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ — bookwork escape hatch if Q6 looks ugly.	07	Q5c	2/4	2/4

UNIT IV — Orthogonal Projections

20 marks · Q7 = 6+7+7 | Q8 = 10+10

Q8 is QR + least-squares — both 4/4, and that's the whole 20. Take it if your algebra holds up. **Q7 is the safer three-problem question** — and the four-subspaces question is *always* in Q7 (4/4).

#	Question — write this	Marks	Section	Topic came up	SAME values
1	QR factorization. Mar 24 used $\begin{bmatrix} 3 & -5 & 1 \\ 1 & 1 & 1 \\ -1 & 5 & -2 \\ 3 & -7 & 8 \end{bmatrix}$. Matrix changes every paper — own the method: Gram-Schmidt the columns, normalise → Q , then $R = Q^T A$.	10	Q8a	4/4	0/4
2	Least-squares solution of $AX = b$: $A = \begin{bmatrix} 1 & 3 & 5 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 3 & 3 \end{bmatrix}$ $b = \begin{bmatrix} 3 \\ 5 \\ 7 \\ -3 \end{bmatrix}$ ← identical values, twice. Solve $A^T A x = A^T b$.	10	Q8b — with #1	4/4	2/4
3	Dimension and basis for the four fundamental subspaces of $A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}$ ← same matrix in 3 of 4 papers. row 3 = row 1 ⇒ rank 2; dim nul = 2, dim left-nul = 1	07	Q7 — always	4/4	3/4
4	Complete solution of $Ax = b$. Mar 24: $x+3y+3z=1$, $2x+6y+9z=5$, $-x-3y+3z=5$. Answer as $x = x_p + c_1 n_1 + \dots$	06	Q7a	3/4	0/4
5	Orthogonal projection. $y = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, $u = \begin{bmatrix} 4 \\ -7 \end{bmatrix}$. Find proj of y onto u ; split y into a part in $\text{span}\{u\}$ and a part orthogonal to it. proj = $((y \cdot u)/(u \cdot u))u$ — 5 minutes, 7 marks	07	Q7b	1/4	1/4
6	Gram-Schmidt orthonormal basis — e.g. $\text{span}\{(1, 1, 1, 1), (0, 1, 1, 1), (0, 0, 1, 1)\}$. You already know this from QR: just stop before forming R .	10	Q8a substitute	1/4	1/4

UNIT V — Applications of Eigenvalue Decomposition

20 marks · a(5) + b(5) + c(10)

PICK the question containing “orthogonally diagonalize” (Q10 in Mar 24). The two property-checks travel with it in 3 of 4 papers — 10 + 5 + 5, and the 5-markers are ten minutes total. SVD sits in the *other* question.

#	Question — write this	Marks	Section	Topic came up	SAME values
1	Orthogonally diagonalize $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$ ← THE trigger. Same matrix, 3 of 4 papers. eigenvalues 2, 3, 6 — all distinct, so eigenvectors come out orthogonal on their own. Normalise → P , then $P^TAP = D$.	10	Q10c	4/4	3/4
2	Is $A = \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix}$ positive definite? minors 1, 0, 0 → positive semi-definite, NOT positive definite. Say so explicitly.	05	Q10a — with #1	4/4	2/4
3	Is $A = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}$ unitary? Show $A^*A = I$ (yes) OR the Hermitian version: is $\begin{bmatrix} 3 & 7-4i & -2+5i \\ 7+4i & -2 & 3+i \\ -2-5i & 3-i & 4 \end{bmatrix}$ Hermitian? (yes)	05	Q10b — with #1, #2	4/4	2/4
4	Find the SVD of $A = \begin{bmatrix} 1 & 1 \\ 3 & -3 \end{bmatrix}$ $A^TA = \begin{bmatrix} 10 & -8 \\ -8 & 10 \end{bmatrix}$, eigenvalues 18 and 2 ⇒ singular values $3\sqrt{2}$ and $\sqrt{2}$	10	OTHER question (Q9c)	4/4	2/4
5	Quadratic form → no cross-product terms. $Q(x) = x_1^2 - 8x_1x_2 - 5x_2^2$. Build the symmetric matrix, orthogonally diagonalise, substitute $x = Py$.	05	Q9a	2/4	0/4
6	Steady state of the Markov matrix $A = \begin{bmatrix} 0.95 & 0.03 \\ 0.05 & 0.97 \end{bmatrix}$ solve $(A-I)x = 0$, scale so entries sum to 1	05	Q9b	1/4	1/4
7	Diagonalize and hence find A^n ($A^n = PD^nP^{-1}$). Appeared 3 papers straight, then skipped in Mar 24 — which makes it overdue , not safe.	10	Q9 (10-mark slot)	3/4	0/4

IF YOU HAVE 20 SPARE MINUTES — the one hole in your prep. PCA is named in your syllabus *and* in CO5, and has **never** been asked in four papers. If it comes, it's 5 marks in Unit V, most likely next to the SVD. All you need: centre the data (subtract the column means), form the covariance matrix $C = \frac{1}{n-1}X^TX$, find its eigenvalues and eigenvectors, take the eigenvector of the largest eigenvalue as the first principal component, and report $\lambda_1/\Sigma\lambda_i$ as the variance it explains. **Conic sections** is the same gap — diagonalise the quadratic form; both eigenvalues the same sign → ellipse, opposite signs → hyperbola.

THE GUARANTEE AUDIT — can this sheet actually clear 50?

short answer: the content floor is 66

What I did. I tagged **every single part of every question in all four papers** by topic — **111 part-slots, 800 marks** — and matched each one against the rows on pages 1–3. Then I computed, for each unit of each paper, how much the better of the two questions would have scored. No estimates.

Audit check	Result
Part-slots examined (4 papers, every sub-part)	111 / 800 marks
Marks covered by the rows on pages 1–3	774 / 800 = 96.8%
Units where <i>at least one</i> of the two questions is 100% covered	20 of 20 (5 units × 4 papers)
March 2024 paper (your current format)	33 / 33 parts — no gaps at all
Topics found that this sheet does not teach	4 — all listed below

The four gaps — and the one line that closes each
Every one of these sits in a question you would never choose — in all 20 unit-paper combinations, the *other* question was fully covered. All four are Unit I/II bookwork. Thirty minutes closes them and takes coverage to 100%.

Gap	Where	Marks	How to close it
Transform of an integral	Apr 23 Q1a(ii) Jun 23 Q2a(ii)	~3 each	One formula: $L\{ \int_0^t f(u) du \} = \frac{F(s)}{s}$. Both papers paired it with a transform you already know.
Prove $L\{f(t)/t\} = \int_s^\infty F(u) du$	Jun 23 Q2b	7	Write $\int_s^\infty F(u) du = \int_s^\infty [\int_0^\infty e^{-ut} f(t) dt] du$, swap the order , integrate in $u \rightarrow e^{-st}/t$. Done.
Prove $L\{(\sinh at)f(t)\} = \frac{1}{2}[F(s-a) - F(s+a)]$	Sep 23 Q2b	7	$\sinh at = \frac{1}{2}(e^{at} - e^{-at})$, then apply the first shifting theorem to each term. Three lines.
State and prove the convolution theorem	Sep 23 Q3a	6	Write $L\{f\} \cdot L\{g\}$ as a double integral, substitute $t = u+v$, swap the order of integration.

THE 14-ITEM CORE — what actually carries you past 50

worst of the four papers: 66/100

These 14 items alone — ignoring QR, SVD, least-squares, Gram-Schmidt, A^n , the improper integrals and everything else on pages 1–3 — would have scored **66, 91, 72 and 86** on the four papers. **The worst case is 66.** If your revision time only stretches to one thing, make it this list.

#	Item	Unit	Marks	Work involved — why it is low-risk
1	t^n-multiplication proof	I	07	Pure recall. No arithmetic at all.
2	$\sin \sqrt{t} \rightarrow \cos \sqrt{t}/\sqrt{t}$	I	04–07	Differentiate a given result. Four lines.
3	The 2-mark definitions (LT, periodic, unit step, Dirac)	I, II	02×2	Four sentences and two sketches. Free marks in <i>both</i> options.
4	Short inverse LT — partial fractions <i>and</i> the log form	II	04–06	One in each question, so it never depends on your choice.
5	Heaviside piecewise + LT	II	07	Mechanical once you have the pattern. Same values twice.
6	ODE / simultaneous ODE by LT	II	07	Always next to #5 (4/4). Longer, but the method never varies.
7	Composition of transformations	III	06	Three 2×2 matrices multiplied. Two minutes.
8	Transition matrix	III	07	One 2×2 inverse. Same bases twice running.
9	Kernel & range + rank-nullity	III	07	One row reduction, then read it off.
10	Four fundamental subspaces	IV	07–10	One row reduction of a matrix that has never changed. Always in Q7.
11	Complete solution of $Ax = b$	IV	06–10	Row reduction again — the same skill as #10.
12	Positive-definite check	V	05	Three determinants. Under two minutes.
13	Unitary / Hermitian check	V	05	One conjugate transpose. Under two minutes.
14	Orthogonal diagonalization	V	10	The one long computation — but the matrix repeated in 3 of 4 papers, so you can rehearse it exactly.

THE HONEST VERDICT. I am not going to tell you this is a certainty, because one part of it is not mine to promise. Here is the split:

- **What I can stand behind:** the *content* is there. 96.8% of 800 marks across four papers, 100% of the current-format paper, and in all 20 unit-paper combinations at least one question was fully covered. The 14-item core alone floors at **66** — a **16-mark cushion** over 50. You could completely blank on orthogonal diagonalization (10) *and* the ODE (7) and still be at 49–50 from the rest.

- **What I cannot promise:** execution. Knowing the method and getting the arithmetic right in three hours are different things, and a sign error in a 10-marker is worth more than any prediction. To land below 50 with this core actually learned, you would have to lose **more than a quarter** of what you attempt to slips.

- **The one structural risk:** choosing the wrong question. Use the triggers — and if a trigger is missing from a unit, do not force it: count the marks you can genuinely do in each of the two questions and take the larger. That takes thirty seconds and protects the whole plan.

How to read this: “Section” is the slot the question occupied in the March 2024 paper (the only one under your current course title) with the older papers cross-checked. “Freq” is how many of the four papers carried that topic. Items marked **identical values** have been set with the same numbers more than once. Predictions are inferences from four papers — strong, not guaranteed. Full 22-page analysis, pairing laws and the complete question index are in *24CS31_Prediction_Sheet.pdf*.